

マクロ計量経済学

第6回 講義ノート マクロ動学モデルの解法 その6

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outline

1. Why 2nd approximation to policy function is needed
2. Set up of the method
3. 1st approximation to policy function
4. 2nd approximation to policy function
 - Matlab code by Schmitt-Grohe & Uribe (2004, JEDC)
5. Example 1: Neo-classical model
 - Matlab Codes
6. Example 2: two-country RBC model
 - Matlab Codes
7. Example 3: Asset price model
 - Matlab code

Reference

1. Miao (2014) chapter 2, 2-4, and 2-5
2. Schmitt-Grohe and Uribe (2004)
“Solving dynamic general equilibrium models using a second-order approximation to the policy function,” JEDC, 28, 755-775.
3. Coeurdacier, Rey, Winant (2011)
“The Risky Steady State,” AER, 101:3, 398-401.
4. De Groot (2013)
“Computing the risky steady state of DSGE models,” Economics Letters, 120, 566-569.

マクロ経済の動学モデルの解法

- **解法その1**: 確率的ベルマン方程式 Stochastic Bellman Equation
→ 応用範囲が広いが、数値的解法で時間がかかる。
 - 状態変数の動学は、マルコフ決定過程
 - ラグランジュ方程式に置き換えることが可能。
- **解法その2**: 合理的期待モデル Rational Expectation Model
⇒ 数値的解法での負担が少ない。
 - Uhlig (1997)
 - Blanchard & Kahn (1980)
 - Schmit-Grohe & Uribe (2004)
- **均衡解の動学式**はマルコフ決定過程
 - 行列Pは 固有値が1以下で安定

$$x_t = Px_{t-1} + Qz_t, \quad \text{均衡解}$$

$$y_t = Rx_{t-1} + Sz_t.$$

合理的期待モデル $E_t(f(y_{t+1}, x_{t+1}))$ に policy function を代入

Many nonlinear rational expectations models can be represented by the following system:

$$E_t [f(\overset{\text{Control}}{\underset{\text{variables}}{y_{t+1}}}, \overset{\text{State}}{\underset{\text{variables}}{y_t}}, x_{t+1}, x_t)] = 0, \quad (2.27)$$

where x_t is an $(n_k + n_z) \times 1$ vector of predetermined variables and y_t is an

$n_y \times 1$ vector of non-predetermined variables in the sense of Blanchard and Kahn (1980). Let $n = n_k + n_z + n_y$ and assume the function $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^n$.

The vector x_t consists of n_k endogenous variables and n_z exogenous shocks:

Suppose that the solution to (2.27) takes the following form:

$$y_t = g(x_t, \sigma), \quad x_{t+1} = h(x_t, \sigma) + \eta \sigma \varepsilon_{t+1}, \quad (2.29)$$

Policy function Dynamics of predetermined variable

where g and h are functions to be determined and η is an $n_x \times n_\varepsilon$ matrix defined by

$$\eta = \begin{bmatrix} 0 \\ \Sigma \end{bmatrix}.$$

合理的期待モデルに1st order approximationを代入

. We totally differentiate (2.27) to obtain the linearized system:

合理的期待モデルの全微分

$$\underset{y'=y_{t+1}}{f_{y'}} E_t dy_{t+1} + f_y dy_t + \underset{x'=x_{t+1}}{f_{x'}} E_t dx_{t+1} + f_x dx_t = 0,$$

where the Jacobian matrix of f is evaluated at the steady state. Adopting the notation introduced in Section 1.6, we can also derive a log-linearized system. Both systems are in the form of (2.11), e.g.,

$$\begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix} \begin{bmatrix} E_t dx_{t+1} \\ E_t dy_{t+1} \end{bmatrix} = - \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} dx_t \\ dy_t \end{bmatrix}. \quad (2.28)$$

We then define

$$0 = F(x, \sigma)$$

$$\equiv E f \left(\underset{=y_{t+1}}{g \left(\underset{=y_t}{h(x, \sigma) + \eta \sigma \varepsilon'}, \sigma \right)}, \underset{=y_t}{g(x, \sigma)}, \underset{=x_{t+1}}{h(x, \sigma) + \eta \sigma \varepsilon'}, x \right) \quad (2.30)$$

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = 0,$$

$$y_t = g(x_t, \sigma), \quad x_{t+1} = h(x_t, \sigma) + \eta \sigma \varepsilon_{t+1},$$

We then use (2.30) to compute

$$\begin{aligned} F_\sigma(\bar{x}, 0) &= E \{ f_{y'} [g_x(h_\sigma + \eta \varepsilon') + g_\sigma] + f_y g_\sigma + f_{x'}(h_\sigma + \eta \varepsilon') \} \\ \text{=dF/d}\sigma &= f_{y'} [g_x h_\sigma + g_\sigma] + f_y g_\sigma + f_{x'} h_\sigma = 0, \end{aligned}$$

where all derivatives are evaluated at the non-stochastic steady state. This is a system of homogeneous linear equations for g_σ and h_σ . For a unique solution to exist, we must have

$$g_\sigma = h_\sigma = 0.$$

$$F_x(\bar{x}, 0) = f_{y'} g_x h_x + f_y g_x + f_{x'} h_x + f_x = 0.$$

=dF/dx

We can equivalently rewrite the above equation as

$$\begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix} h_x = - \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} I \\ g_x \end{bmatrix}. \quad (2.31)$$

Let $A = \begin{bmatrix} f_{x'} & f_{y'} \end{bmatrix}$, $B = - \begin{bmatrix} f_x & f_y \end{bmatrix}$, and $\hat{x}_t = x_t - \bar{x}$. Postmultiplying $\hat{x}_t$ on each side of (2.31) yields:

$$A \begin{bmatrix} I \\ g_x \end{bmatrix} h_x \hat{x}_t = B \begin{bmatrix} I \\ g_x \end{bmatrix} \hat{x}_t. \quad (2.32)$$

Using the fact that $\hat{x}_{t+1} = h_x \hat{x}_t$ and $\hat{y}_t = g_x \hat{x}_t$ for the linearized non-stochastic solution, we obtain

$$A \begin{bmatrix} \hat{x}_{t+1} \\ \hat{y}_{t+1} \end{bmatrix} = B \begin{bmatrix} \hat{x}_t \\ \hat{y}_t \end{bmatrix}. \quad (2.33)$$

1次近似(2.33)式は Blanchard-Kahnのモデルに変換できる

$$A \begin{bmatrix} \hat{x}_{t+1} \\ \hat{y}_{t+1} \end{bmatrix} = B \begin{bmatrix} \hat{x}_t \\ \hat{y}_t \end{bmatrix}. \quad (2.33)$$

Blanchard & Kahn (1980)

6.8.1 General Version

A linear model can be written (in what is known as a state space representation) as

$$\begin{matrix} \text{State variables} \\ B \begin{bmatrix} x_{t+1} \\ E_t y_{t+1} \end{bmatrix} = A \begin{bmatrix} x_t \\ y_t \end{bmatrix} + G \varepsilon_t, \end{matrix} \quad (6.13)$$

Control variables

where x_t is an $(n \times 1)$ vector of predetermined variables at date t , y_t is an $(m \times 1)$ vector of non-predetermined variables at time t , $E_t y_{t+1}$ is the $(m \times 1)$ vector of expectations for the non-predetermined variables at date $t + 1$, ε_t is a $(k \times 1)$ vector of stochastic shocks, A and B are $((n + m) \times (n + m))$ matrices, and G is an $((n + m) \times k)$ matrix. The difference between predetermined and non-predetermined variables is that the values of the predetermined variables at time $t + 1$ do not depend on the values of the time $t + 1$ shocks, while the values of the non-predetermined variables do depend on them. That is why, at

2nd order approximation

Our goal is to find an approximate solution for g and h . We shall consider local approximation around a particular point $(\bar{x}, \bar{\sigma})$. In applications, a convenient point to take is the non-stochastic steady state $(\bar{x}, \bar{\sigma})$. At this point, $f(\bar{y}, \bar{y}, \bar{x}, \bar{x}) = 0$. We then write the Taylor approximation up to the second order,

$$\begin{aligned} g(x, \sigma) = & g(\bar{x}, 0) + g_x(\bar{x}, 0)(x - \bar{x}) + g_\sigma(\bar{x}, 0)\sigma \\ & + \frac{1}{2}g_{xx}(\bar{x}, 0)(x - \bar{x})^2 + g_{x\sigma}(\bar{x}, 0)(x - \bar{x})\sigma \\ & + \frac{1}{2}g_{\sigma\sigma}(\bar{x}, 0)\sigma^2, \end{aligned}$$

$$\begin{aligned} h(x, \sigma) = & h(\bar{x}, 0) + h_x(\bar{x}, 0)(x - \bar{x}) + h_\sigma(\bar{x}, 0)\sigma \\ & + \frac{1}{2}h_{xx}(\bar{x}, 0)(x - \bar{x})^2 + h_{x\sigma}(\bar{x}, 0)(x - \bar{x})\sigma \\ & + \frac{1}{2}h_{\sigma\sigma}(\bar{x}, 0)\sigma^2, \end{aligned}$$

We then define

$$\begin{aligned}
 0 &= F(x, \sigma) \\
 &\equiv Ef \left(\underset{=y_{t+1}}{g(h(x, \sigma) + \eta\sigma\varepsilon', \sigma)}, \underset{=x_{t+1}}{g(x, \sigma)}, h(x, \sigma) + \eta\sigma\varepsilon', x \right) \quad (2.30)
 \end{aligned}$$

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = 0,$$

$$y_t = g(x_t, \sigma), \quad x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1},$$

We compute

$$\begin{aligned}
 0 &= F_{xx}(\bar{x}, 0) = [f_{y'y}g_x h_x + f_{y'y}g_x \\
 &\quad + f_{y'x'}h_x + f_{y'x}]g_x h_x n_x \\
 &\quad + f_{y'}g_{xx}h_x h_x + f_{y'}g_{xx}h_{xx} n_x \times n_x \\
 &\quad + [f_{yy'}g_x h_x + f_{yy}g_x + f_{yx'}h_x + f_{yx}]g_x + f_y g_{xx} n_y \times n_x \\
 &\quad + [f_{x'y'}g_x h_x + f_{x'y}g_x + f_{x'x'}h_x + f_{x'x}]h_x + f_{x'}h_{xx} n_x \times n_x \\
 &\quad + f_{xy'}g_x h_x + f_{xy}g_x + f_{xx'}h_x + f_{xx}.
 \end{aligned}$$

Since we know the derivatives of f as well as the first derivatives of g and h evaluated at $(\bar{y}, \bar{y}, x, x)$, it follows that the above equation represents a system of $n \times n_x \times n_x$ linear equations for the same number of unknowns given by the elements of g_{xx} and h_{xx} .

Similarly, we compute $g_{\sigma\sigma}$ and $h_{\sigma\sigma}$ by solving the equation $F_{\sigma\sigma}(\bar{x}, 0) = 0$.

We compute

$$\begin{aligned} 0 &= F_{\sigma\sigma}(\bar{x}, 0) = f_{y'}g_x h_{\sigma\sigma} + f_{y'y'}g_x \eta g_x \eta I \\ &\quad + f_{y'x'}\eta g_x \eta I + f_{y'}g_{xx}\eta \eta I + f_{y'}g_{\sigma\sigma} \\ &\quad + f_y g_{\sigma\sigma} + f_{x'}h_{\sigma\sigma} \\ &\quad + f_{x'y'}g_x \eta \eta I + f_{x'x'}\eta \eta I. \end{aligned}$$

This is a system of n linear equations for n unknowns $g_{\sigma\sigma}$ and $h_{\sigma\sigma}$.

Finally, we show that the cross derivatives $g_{\sigma x}$ and $h_{\sigma x}$ are equal to zero when evaluated at $(\bar{x}, 0)$. We compute

$$0 = F_{\sigma x}(\bar{x}, 0) = f_{y'}g_x h_{\sigma x} + f_{y'}g_{\sigma x}h_x + f_y g_{\sigma x} + f_{x'}h_{\sigma x}.$$

This is a system of $n \times n_x$ linear homogenous equations for $n \times n_x$ unknowns $g_{\sigma x}$ and $h_{\sigma x}$. If a unique solution exists, then the solution must be equal to zero.

Matlab codes of 2nd Approximation of policy function (Smitt-Grohe and Uribe, 2004, JEDC)

Matlab code

First-order approximation

[gx hx.m](#)

Second-order approximation

[gxx hxx.m](#)

[gss hss.m](#)

Obtaining the derivatives of f

(requires Matlab's Symbolic Math Toolbox)

[anal_deriv.m](#)

[num_eval.m](#)

Model Simulation:

[simu_2nd.m](#)

Writing the Output of `anal_deriv.m` to an M-file: This program saves significant amount of computational time in applications in which the output of `anal_deriv.m` has to be evaluated repeatedly.

[anal_deriv_print2f.m](#) by [Andrea Pescatori](#)

Example 1: The neoclassical growth model: **Contents of Matlab Codes**

- neoclassical_model_run.m → main program
- neoclassical_model.m →
description of model equations (FOC and constraints)
- neoclassical_model_ss →
set up of parameters
and (non-stochastic) Steady State
- neoclassical_model_simul →
calculation of IRF, Artificial data (Monte Carlo)
and stochastic Steady State

Main code その1

```
26
27 % analytical 2nd order derivatives of system: f
28 [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,fyy,...
29  fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxxy,fxxp,fx,x,f] = neoclassical_model;
30
31 %% Numerical Evaluation
32 % Non-stochastic Steady State and Parameter Values
33 [SIG,DELTA,ALFA,BETTA,RHO,eta,c,cp,k,kp,a,ap,A,K,C]=neoclassical_model_ss;
34
35 %Order of approximation desired
36 approx = 2;
37
38 %Obtain numerical derivatives of f
39 num_eval
40
41 %First-order approximation of Policy function
42 [gx,hx] = gx_hx(nfy,nfx,nfyp,nfxp)
```

Main code その2

```
41 %First-order approximation of Policy function
42 [gx,hx] = gx_hx(nfy,nfx,nfyp,nfxp)
43
44 %Second-order approximation of Policy function
45 [gxx,hxx] = gxx_hxx(nfx,nfxp,nfy,nfyp,nfypyp,nfypy,nfypxp,...
46                    nfypx,nfyyp,nfyy,nfyxp,nfyx,nfxpyp,...
47                    nfxpy,nfxpxp,nfxpx,nfxyp,nfxy,nfxxp,nfxx,hx,gx)
48
49 [gss,hss] = gss_hss(nfx,nfxp,nfy,nfyp,nfypyp,nfypy,nfypxp,nfypx,...
50                    nfyyyp,nfyy,nfyxp,nfyx,nfxpyp,nfxpy,nfxpxp,nfxpx,...
51                    nfxyp,nfxy,nfxxp,nfxx,hx,gx,gxx,eta)
52
53
54 %% Simulation of solution
55 neoclassical_model_simul;
56
```

Example 1: The neoclassical growth model

Sec 2 of Smitt-Grohe and Uribe (2004)

$$c_t^{-\gamma} = \beta E_t c_{t+1}^{-\gamma} [\alpha A_{t+1} k_{t+1}^{\alpha-1} + 1 - \delta], \quad \text{Euler equation}$$

$$c_t + k_{t+1} = A_t k_t^\alpha + (1 - \delta) k_t, \quad \text{Dynamics of capital}$$

$$\ln A_{t+1} = \rho \ln A_t + \sigma \varepsilon_{t+1}, \quad \text{TFP shock process}$$

for all $t \geq 0$, given k_0 and A_0 . Let $y_t = c_t$ and $x_t = [k_t; \ln A_t]'$. Then

$$E_t f(y_{t+1}, y_t, x_{t+1}, x_t) = E_t \begin{bmatrix} y_{1t}^{-\gamma} - \beta y_{1t+1}^{-\gamma} [\alpha e^{x_{2t+1}} x_{1t+1}^{\alpha-1} + 1 - \delta] \\ y_{1t} + x_{1t+1} - e^{x_{2t}} x_{1t}^\alpha - (1 - \delta) x_{1t} \\ x_{2t+1} - \rho x_{2t} \end{bmatrix},$$

Neoclassical_model.m

```
13  %(c) Stephanie Schmitt-Grohe and Martin Uribe
14  %Date July 17, 2001, revised 22-Oct-2004
15
16  %Define parameters
17  syms    SIG DELTA ALFA BETTA RHO
18
19  %Define variables
20  syms c cp k kp a ap
21
22  %Write equations fi, i=1:3
23  f1 = c + kp - (1-DELTA) * k - a * k^ALFA;
24  f2 = c^(-SIG) - BETTA * cp^(-SIG) * (ap * ALFA * kp^(ALFA-1) + 1 - DELTA);
25  f3 = log(ap) - RHO * log(a);
26
27  %Create function f
28  f = [f1;f2;f3];
```

define:

$$x_t = \begin{bmatrix} \ln k_t \\ \ln A_t \end{bmatrix} : \text{Predetermined variables}$$

and

$$y_t = \ln c_t. \quad : \text{Control variable}$$

Then the non-stochastic steady-state values of y_t and x_t are, respectively:

$$\bar{y} = -0.8734.$$

and

$$\bar{x} = \begin{bmatrix} -1.7932 \\ 0 \end{bmatrix}.$$

```
29
30 % Define the vector
31 x = [k a];
32 y = c;
33 xp = [kp ap];
34 yp = cp;
35
```

`function [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fx,x,f] = neoclassical_model`

```
35
36 %Make f a function of the logarithm of the state and contr
37 f = subs(f, [x,y,xp,yp], (exp([x,y,xp,yp])));
38 %if line 36 gives an error (whichh it will for some versic
39 %f = subs(f, [x,y,xp,yp], transpose(exp([x,y,xp,yp])));
40
41 %Compute analytical derivatives of f
42 approx = 2; %1:1st order, 2: 2nd order
43 [fx,fxp,fy,fyp,fypyp,fypy,fypxp,fypx,fyyp,...
44 fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,...
45 fxy,fxxp,fx,x] = anal_deriv(f,x,y,xp,yp,approx);
46
```

このページの内容は最新ではありません。最新版の英語を参照す

subs

シンボリック代入

構文

```
snew = subs(s,old,new)
snew = subs(s,new)
snew = subs(s)
```

```
sMnew = subs(sM,oldM,newM)
sMnew = subs(sM,newM)
sMnew = subs(sM)
```

例

へ 単一置換

この式で a を 4 に置き換えます。

```
syms a b
subs(a + b, a, 4)
```

$ans = b + 4$

この式で $a*b$ を 5 に置き換えます。

```
subs(a*b^2, a*b, 5)
```

$ans = 5b$

```
function [fx,fxp,fy,fyp,fypyp,fypy,fypxp,...
fypx,fyyp,fyy,fyxp,fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxy,fxxp,fxx]...
=anal_deriv(f,x,y,xp,yp,approx);
```

```

5  if nargin==5 %5つ目の入力为空欄のケース
6  approx=2;
7  end
8
9  nx = size(x,2);
10 ny = size(y,2);
11 nxp = size(xp,2);
12 nyp = size(yp,2);
13
14 n = size(f,1);
15
16 %Compute the first and second derivatives of f
17 fx = jacobian(f,x);
18 fxp = jacobian(f,xp);
19 fy = jacobian(f,y);
20 fyp = jacobian(f,yp);
21
```

```

a 22  if approx==2
a 23  fypyp = reshape(jacobian(fyp(:),yp),n,nyp,nyp);
24  fypy = reshape(jacobian(fyp(:),y),n,nyp,ny);
25  fypxp = reshape(jacobian(fyp(:),xp),n,nyp,nxp);
26  fypx = reshape(jacobian(fyp(:),x),n,nyp,nx);
27  fyyp = reshape(jacobian(fy(:),yp),n,ny,nyp);
28  fyy = reshape(jacobian(fy(:),y),n,ny,ny);
29  fyxp = reshape(jacobian(fy(:),xp),n,ny,nxp);
30  fyx = reshape(jacobian(fy(:),x),n,ny,nx);
31  fxpyp = reshape(jacobian(fxp(:),yp),n,nxp,nyp);
32  fxpy = reshape(jacobian(fxp(:),y),n,nxp,ny);
33  fxpxp = reshape(jacobian(fxp(:),xp),n,nxp,nxp);
34  fxpx = reshape(jacobian(fxp(:),x),n,nxp,nx);
35  fxyp = reshape(jacobian(fx(:),yp),n,nx,nyp);
36  fxy = reshape(jacobian(fx(:),y),n,nx,ny);
37  fxxp = reshape(jacobian(fx(:),xp),n,nx,nxp);
38  fxx = reshape(jacobian(fx(:),x),n,nx,nx);
39  else
40  fypyp=0; fypy=0; fypxp=0; fypx=0; fyyp=0; fyy=0;...
41  fyxp=0; fyx=0; fxpyp=0; fxpy=0; fxpxp=0;...
42  fxpx=0; fxyp=0; fxy=0; fxxp=0; fxx=0;
43  end
```

reshape

既存の要素を再配列して配列を形状変更する

R2024b

ページ内をすべて折りたたむ

構文

```
B = reshape(A,sz)
B = reshape(A,sz1,...,szN)
```

展開

例

へ ベクトルを行列に形状変更

1 行 10 列のベクトルを 5 行 2 列の行列に形状変更します。

```
A = 1:10;
B = reshape(A,[5,2])
```

B = 5×2

1	6
2	7
3	8
4	9
5	10

1次近似 gx_hx()

```
1 function [gx,hx,exitflag]=gx_hx(fy,fx,fyp,fxp,staking);
2 |
3 if nargin<5
4     stake=1;
5 end
6 exitflag = 1;
7
8 %Create system matrices A,B
9 A = [-fxp -fyp];
0 B = [fx fy];
1 NK = size(fx,2);
2
3 %Complex Schur Decomposition
4 [s,t,q,z] = qz(A,B);
5
6 %Pick non-explosive (stable) eigenvalues
7 slt = (abs(diag(t))<stake*abs(diag(s)));
8 nk=sum(slt);
9
0 %Reorder the system with stable eigs in upper-left
1 [s,t,q,z] = ordqz(s,t,q,z,slt);
2
3 %Split up the results appropriately
4 z21 = z(nk+1:end,1:nk);
5 z11 = z(1:nk,1:nk);
6
7 s11 = s(1:nk,1:nk);
8 t11 = t(1:nk,1:nk);
9
```

```
29
30 %Identify cases with no/multiple solutions
31 if nk>NK
32     warning('The Equilibrium is Locally Indeterminate')
33     exitflag=2;
34 elseif nk<NK
35     warning('No Local Equilibrium Exists')
36     exitflag = 0;
37 end
38
39 if rank(z11)<nk;
40     warning('Invertibility condition violated')
41     exitflag = 3;
42 end
43
44 %Compute the Solution
45 z11i = z11\eye(nk);
46 gx = real(z21*z11i);
47 hx = real(z11*(s11\t11)*z11i);
48
```

2次近似

```
23
24 function [gxx,hxx] = gxx_hxx(fx,fxp,fy,fyp, ...
25     fypyp,fypy,fypxp,fypx,fyyp,fyy,fyxp, ...
26     fyx,fxpyp,fxpy,fxpxp,fxpx,fxyp,fxxy,fxxp,fxx,hx,gx)
27
28 m=0;
29 nx = size(hx,1); %rows of hx and hxx
30 ny = size(gx,1); %rows of gx and gxx
31 n = nx + ny; %length of f
32 ngxx = nx^2*ny; %elements of gxx
33
34 sg = [ny nx nx]; %size of gxx
35 sh = [nx nx nx]; %size of hxx
36
37 Q = zeros(n*n*(nx+1)/2,n*n*n);
38 q = zeros(n*n*(nx+1)/2,1);
39 gxx=zeros(sg);
40 hxx=zeros(sh);
41 GXX=zeros(sg);
42 HXX=zeros(sh);
43
```

```
45 for i=1:n
46     for j=1:nx
47         %for k=1:nx
48         for k=1:j
49             m = m+1;
50
51             %First Term
52             q(m,1) = ( shiftdim(fypyp(i,:,:),1) * gx * hx(:,k) ...
53                 + shiftdim(fypy(i,:,:),1) * gx(:,k) + shiftdim(fypxp(i,:,:),1) ...
54                 * hx(:,k) + shiftdim(fypx(i,:,k),1) )' * gx * hx(:,j));
55
56             % Second term
57             GXX(:) = kron(ones(nx^2,1),fyp(i,:))';
58             pGXX = permute(GXX,[2 3 1]);
59             pGXX(:) = pGXX(:) .* kron(ones(nx*ny,1),hx(:,j));
60             GXX=ipermute(pGXX,[2 3 1]);
61             pGXX = permute(GXX,[3 1 2]);
62             pGXX(:) = pGXX(:) .* kron(ones(nx*ny,1),hx(:,k));
63             GXX=ipermute(pGXX,[3 1 2]);
64             Q(m,1:ngxx)=GXX(:)';
65             GXX=0*GXX;
```

```

67 %Third term
68 HXX(:,j,k) = (fyp(i,:) * gx)';
69 Q(m,ngxx+1:end)=HXX(:)';
70 HXX = 0*HXX;
71
72 %Fourth Term
73 q(m,1) = q(m,1) + ( shiftdim(fyyp(i,:,:),1) * gx * hx(:,k) ...
74     + shiftdim(fyy(i,:,:),1) * gx(:,k) + shiftdim(fyxp(i,:,:),1) ...
75     * hx(:,k) + shiftdim(fyx(i,:,k),1) )' * gx(:,j);
76
77 %Fifth Term
78 GXX(:,j,k)=fy(i,:)';
79 Q(m,1:ngxx) = Q(m,1:ngxx) + GXX(:)';
80 GXX = 0*GXX;
81
82 %Sixth term
83 q(m,1) = q(m,1) + ( shiftdim(fxpyp(i,:,:),1) ...
84     * gx * hx(:,k) + shiftdim(fxpy(i,:,:),1) ...
85     * gx(:,k) + shiftdim(fpxp(i,:,:),1) ...
86     * hx(:,k) + fxpx(i,:,k)')' * hx(:,j);
87
88 %Seventh Term
89 HXX(:,j,k)=fxp(i,:)';
90 Q(m,ngxx+1:end) = Q(m,ngxx+1:end) + HXX(:)';
91 HXX = 0*HXX;
92

```

```

92
93 %Eighth Term
94 q(m,1) = q(m,1) + shiftdim(fxyp(i,j,:),1)
95 * gx * hx(:,k) + shiftdim(fxy(i,j,:),1)
96 * gx(:,k) + shiftdim(fxyp(i,j,:),1) * hx(:,k) + fxx(i,j,k);
97
98 end %k
99 end %j
100 end %i
101
102 A = temp(nx,ny);
103
104 Qt = Q* A;
105
106 xt = -Qt\q;
107 x = A* xt;
108
109 gxx(:)=x(1:ngxx);
110 hxx(:) = x(ngxx+1:end);
111

```

The coefficients of the linear terms are:

$$g_x = [0.2525 \quad 0.8417]$$

and

$$h_x = \begin{bmatrix} 0.4191 & 1.3970 \\ 0.0000 & 0.0000 \end{bmatrix}.$$

The coefficients of the quadratic terms are given by:

$$g_{xx}(:, :, 1) = [-0.0051 \quad -0.0171],$$

$$g_{xx}(:, :, 2) = [-0.0171 \quad -0.0569]$$

and

$$h_{xx}(:, :, 1) = \begin{bmatrix} -0.0070 & -0.0233 \\ 0 & 0 \end{bmatrix},$$

$$h_{xx}(:, :, 2) = \begin{bmatrix} -0.0233 & -0.0778 \\ 0 & 0 \end{bmatrix}.$$

Finally, the coefficients of the quadratic terms in σ are:

$$g_{\sigma\sigma} = -0.1921$$

and

$$h_{\sigma\sigma} = \begin{bmatrix} 0.4820 \\ 0 \end{bmatrix}.$$

$$\begin{aligned} g(x, \sigma) = & g(\bar{x}, 0) + g_x(\bar{x}, 0)(x - \bar{x}) + g_\sigma(\bar{x}, 0)\sigma \\ & + \frac{1}{2}g_{xx}(\bar{x}, 0)(x - \bar{x})^2 + g_{x\sigma}(\bar{x}, 0)(x - \bar{x})\sigma \\ & + \frac{1}{2}g_{\sigma\sigma}(\bar{x}, 0)\sigma^2, \end{aligned}$$

Matlabの計算結果

```
gx =
```

```
    0.2525    0.8417
```

```
hx =
```

```
    0.4191    1.3970  
         0         0
```

```
gxx(:, :, 1) =
```

```
   -0.0051   -0.0171
```

```
gxx(:, :, 2) =
```

```
   -0.0171   -0.0569
```

```
hxx(:, :, 1) =
```

```
   -0.0070   -0.0233  
         0         0
```

```
hxx(:, :, 2) =
```

```
   -0.0233   -0.0778  
         0         0
```

```
gss =
```

```
   -0.1921
```

```
hss =
```

```
    0.4820  
         0
```

Policy function

$$y_t = g(x_t, \sigma),$$

Then, the laws of motion of these two variables are given by

$$\hat{c}_t = 0.2525\hat{k}_t + 0.8417\hat{A}_t + \frac{1}{2}[-0.0051\hat{k}_t^2 - 0.0341\hat{k}_t\hat{A}_t - 0.0569\hat{A}_t^2 - 0.1921\sigma^2]$$

and

$$\hat{k}_{t+1} = 0.4191\hat{k}_t + 1.3970\hat{A}_t$$

$$x_{t+1} = h(x_t, \sigma) + \eta\sigma\varepsilon_{t+1},$$

$$+ \frac{1}{2}[-0.0070\hat{k}_t^2 - 0.0467\hat{k}_t\hat{A}_t - 0.0778\hat{A}_t^2 + 0.4820\sigma^2].$$

It can be verified that these numbers coincide with those obtained by Sims (2000).

Neoclassical_simul.m

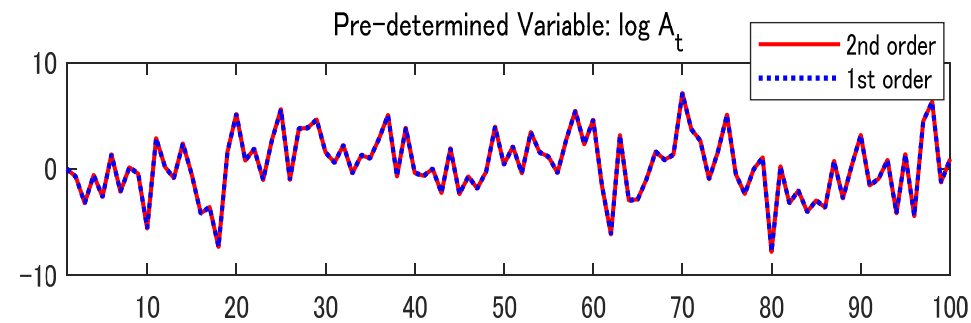
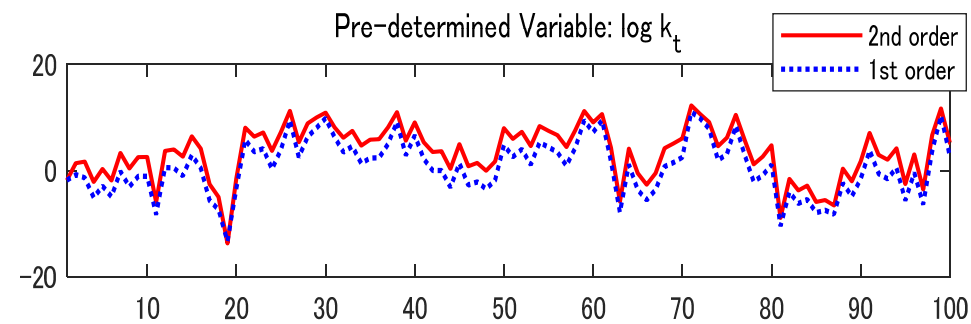
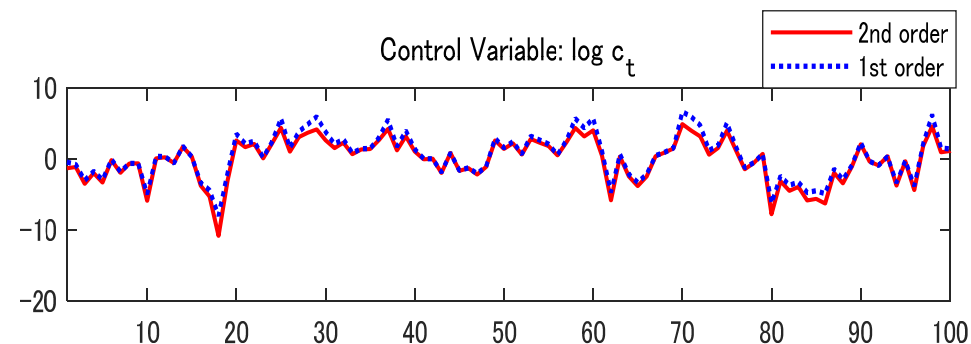
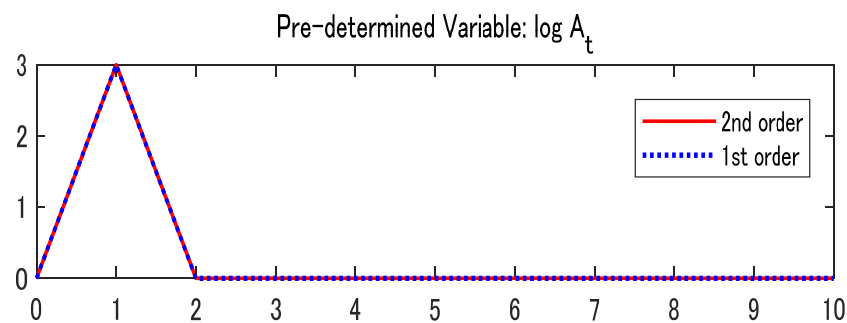
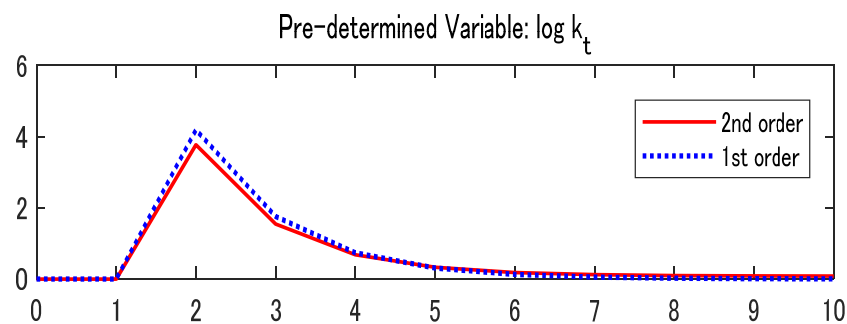
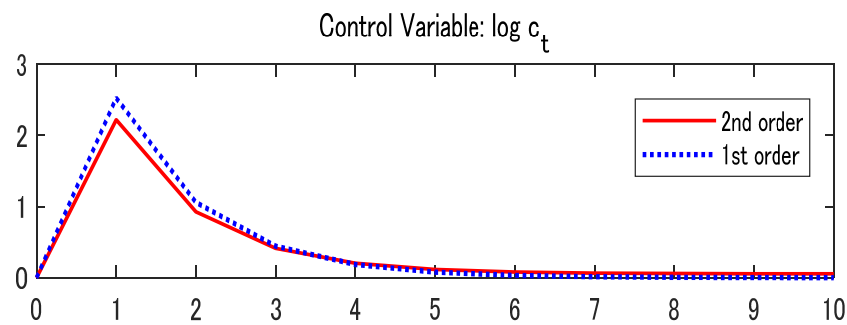
```
1  %% Check Solution
2  % cal unconditional_mean
3  sig = 1; % St Dev of shock
4  [Ey,Ex] = unconditional_mean(gx, hx, gxx, hxx, gss, hss, eta, sig)
5
6  x0 = Ex;
7  e = [ 0 0; zeros(200,2)];
8  [Y02,X02] = simu_2nd(gx, hx, gxx, hxx, gss, hss, eta, sig, x0, e);
9  x_ss1 = X02(end,:)' % stochastic steady state
10
11 % cal stochastic steady state by fixed point
12 x_d = zeros(4,1); % x_d = zeros(4*2,1);
13 x0_ss = csolve('eq_stochastic_steady_state',x_d,[],1e-5,2000,hx, hxx, hss, sig )
14 % x0_ss2 = x0_ss(1:4,1)+x0_ss(5:8,1)
15 [Error] = eq_stochastic_steady_state(x0_ss , hx, hxx, hss, sig )
16
```

```

18 % Impulse Response Function
19 % sig = 0.1;           % St Dev of shock
20 x0_1 = [0;0;0;0];      % initial values of predetermined variables
21 x0_2 = x_ss1;
22 h = 40;                % horizon of IRF
23 e = [ 1 0; zeros(h,2)]; % generations of shocks
24 t = 0:1:h+1;
25
26 [Y2,X2] = simu_2nd(gx, hx, gxx, hxx, gss, hss, eta, sig, x0_2, e);
27 [Y1,X1] = simu_1st(gx, hx, eta, sig, x0_1, e);
28
29 figure('Name','IRF of Kim Model')
30 subplot(3,1,1)
31     plot(t,Y2-Y2(1,1),'r-','LineWidth',1.5);
32     hold on
33     plot(t,Y1-Y1(1,1),'b:','LineWidth',1.5);
34     hold off
35     title('Control Variable: log c_t')
36     legend('2nd order', '1st order')
37     xlim([0 h])
38 %     ylim( [-0.025 0.025])

```

インパルス応答の1次近似と2次近似の比較



Example 2: A two-country neoclassical model: **Contents of Matlab Codes**

- kim_model_run.m → main program
- kim_model.m →
description of model equations (FOC and constraints)
- kim_model_ss →
set up of parameters
and (non-stochastic) Steady State
- kim_model_simul →
calculation of IRF, Artificial data (Monte Carlo)
and stochastic Steady State

Example 2: A two-country neoclassical model with complete asset markets

The planner's objective function is given by

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_{1t}^{1-\gamma} - 1}{1-\gamma} + \frac{C_{2t}^{1-\gamma} - 1}{1-\gamma} \right],$$

where C_{it} , $i=1,2$, denotes consumption of the representative household of country i in period t . The planner maximizes this utility function subject to the budget constraint

$$C_{1t} + C_{2t} + K_{1t+1} - (1-\delta)K_{1t} + K_{2t+1} - (1-\delta)K_{2t} = A_{1t}K_{1t}^{\alpha} + A_{2t}K_{2t}^{\alpha},$$

where K_{it} denotes the stock of physical capital in country i and A_{it} is an exogenous technology shock whose law of motion is given by

$$\ln A_{it} = \rho_i \ln A_{it-1} + \sigma \varepsilon_{it}, \quad i = 1, 2,$$

where $\varepsilon_{it} \sim NIID(0, 1)$ and $\rho_i \in (-1, 1)$. The optimality conditions associated with this problem are the above period-by-period budget constraint and

$$C_{1t} = C_{2t},$$

$$C_{1t}^{-\gamma} = \beta E_t C_{1t+1}^{-\gamma} [\alpha A_{1t+1} K_{1t+1}^{\alpha-1} + (1-\delta)],$$

$$C_{1t}^{-\gamma} = \beta E_t C_{1t+1}^{-\gamma} [\alpha A_{2t+1} K_{2t+1}^{\alpha-1} + (1-\delta)].$$

kim_model.m

```
5
6 %Define parameters
7 syms GAMA DELTA ALFA RHO BETTA
8
9 %Define variables
10 syms c cp k1 k1p k2 k2p a1 a1p a2 a2p
11 %Write equations (e1 e2 e3 e4 e5)
12 e1 = c^(-GAMA) - BETTA * cp^(-GAMA) * (ALFA * a1p * k1p^(ALFA-1) + 1 - DELTA);
13
14 e2 = c^(-GAMA) - BETTA * cp^(-GAMA) * (ALFA * a2p * k2p^(ALFA-1) + 1 - DELTA);
15
16 e3 = 2 * c + k1p - (1-DELTA) * k1 + k2p - (1-DELTA) * k2 - a1 * k1^(ALFA) - a2 * k2^(ALFA);
17
18 e4 = log(a1p) - RHO * log(a1);
19
20 e5 = log(a2p) - RHO * log(a2);
21
22 %Create function f
23 f = [e1;e2;e3;e4;e5];
24
```

We use the following parameter values: $\gamma = 2$; $\delta = 0.1$; $\alpha = 0.3$; $\rho = 0$; and $\beta = 0.95$.

Given this parameterization, the second-order approximation to the policy function is given by:

$$\begin{aligned} \hat{K}_{1t+1} = & [0.4440 \quad 0.4440 \quad 0.2146 \quad 0.2146] \begin{bmatrix} \hat{K}_{1t} \\ \hat{K}_{2t} \\ \hat{A}_{1t} \\ \hat{A}_{2t} \end{bmatrix} \\ & + \frac{1}{2} [\hat{K}_{1t} \quad \hat{K}_{2t} \quad \hat{A}_{1t} \quad \hat{A}_{2t}] \\ & \times \begin{bmatrix} 0.22 & -0.18 & -0.023 & -0.088 \\ -0.18 & 0.22 & -0.088 & -0.023 \\ -0.023 & -0.088 & 0.17 & -0.042 \\ -0.088 & -0.023 & -0.042 & 0.17 \end{bmatrix} \begin{bmatrix} \hat{K}_{1t} \\ \hat{K}_{2t} \\ \hat{A}_{1t} \\ \hat{A}_{2t} \end{bmatrix} \\ & - \frac{1}{2} 0.166 \sigma^2, \end{aligned}$$

$$\hat{K}_{2t+1} = \hat{K}_{1t+1},$$

$$\begin{aligned} [h(x, \sigma)]^j = & [h(\bar{x}, 0)]^j + [h_x(\bar{x}, 0)]_a^j [(x - \bar{x})]_a + [h_\sigma(\bar{x}, 0)]^j [\sigma] \\ & + \frac{1}{2} [h_{xx}(\bar{x}, 0)]_{ab}^j [(x - \bar{x})]_a [(x - \bar{x})]_b \\ & + \frac{1}{2} [h_{x\sigma}(\bar{x}, 0)]_a^j [(x - \bar{x})]_a [\sigma] \\ & + \frac{1}{2} [h_{\sigma x}(\bar{x}, 0)]_a^j [(x - \bar{x})]_a [\sigma] \\ & + \frac{1}{2} [h_{\sigma\sigma}(\bar{x}, 0)]^j [\sigma] [\sigma], \end{aligned}$$

gx =

0.2013	0.2013	0.0973	0.0973
--------	--------	--------	--------

hx =

0.4440	0.4440	0.2146	0.2146
0.4440	0.4440	0.2146	0.2146
0	0	0	0
0	0	0	0

gxx(:, :, 1) =

0.1013	-0.0796	-0.0093	-0.0385
--------	---------	---------	---------

gxx(:, :, 2) =

-0.0796	0.1013	-0.0385	-0.0093
---------	--------	---------	---------

gxx(:, :, 3) =

-0.0093	-0.0385	0.0787	-0.0186
---------	---------	--------	---------

gxx(:, :, 4) =

-0.0385	-0.0093	-0.0186	0.0787
---------	---------	---------	--------

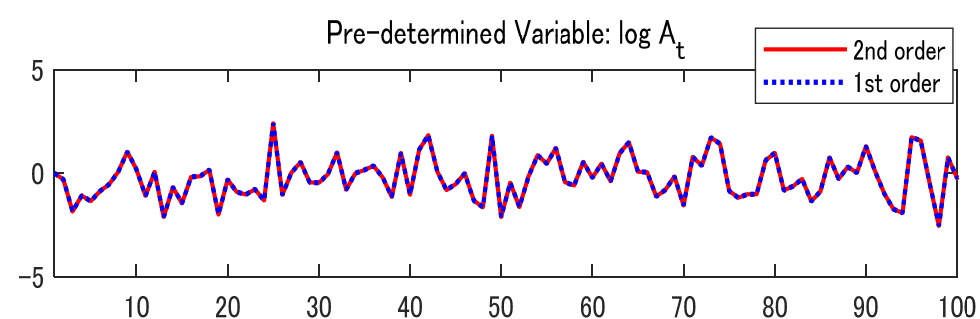
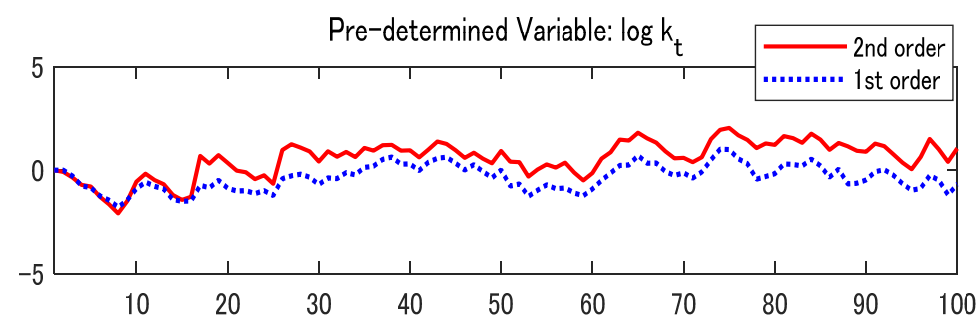
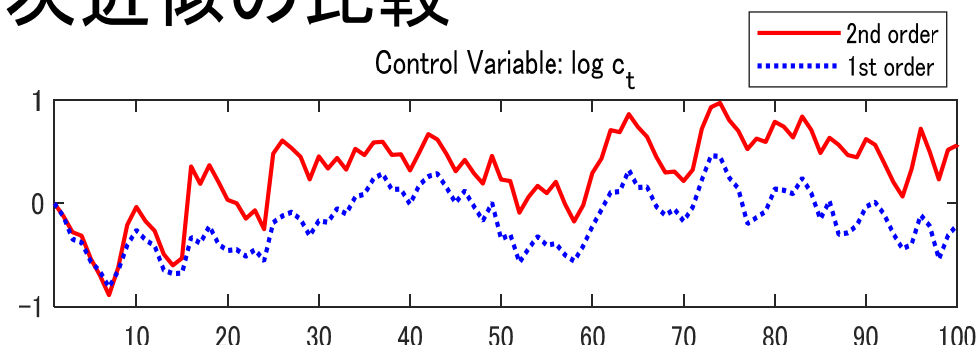
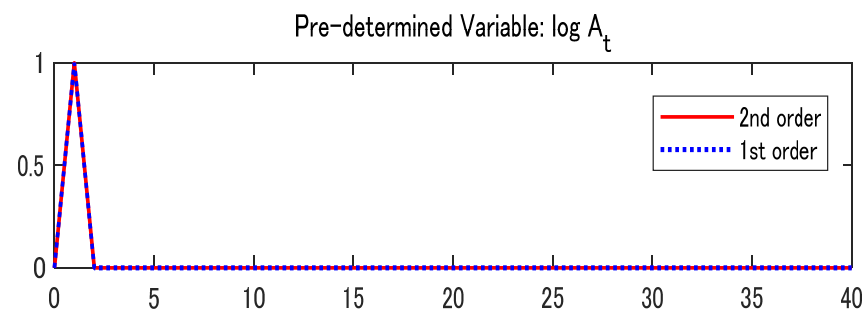
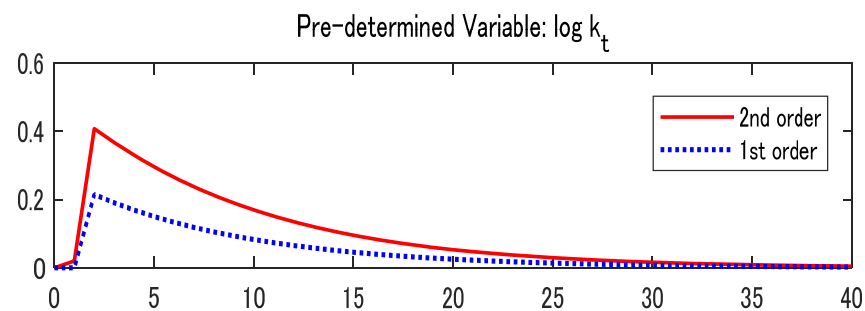
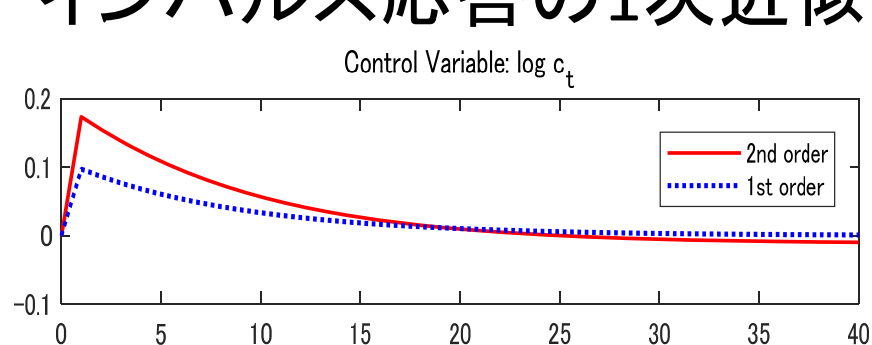
hxx(:, :, 1) =

0.2178	-0.1812	-0.0232	-0.0876
0.2178	-0.1812	-0.0232	-0.0876
0	0	0	0
0	0	0	0

hxx(:, :, 2) =

-0.1812	0.2178	-0.0876	-0.0232
-0.1812	0.2178	-0.0876	-0.0232
0	0	0	0
0	0	0	0

インパルス応答の1次近似と2次近似の比較



Example 3: An asset pricing model: **Contents of Matlab Codes**

- `asset_run.m` → main program
- `asset_model.m` →
description of model equations (FOC and constraints)
- `asset_ss` →
set up of parameters
and (non-stochastic) Steady State
- `asset_model_simul` →
calculation of IRF, Artificial data (Monte Carlo)
and stochastic Steady State

Example 3: An asset pricing model

Consider the following endowment economy analyzed by Burnside (1998) and Collard and Juillard (2001b). The representative agent maximizes the lifetime utility function

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^\theta}{\theta},$$

subject to

$$p_t e_{t+1} + C_t = p_t e_t + d_t e_t$$

and a borrowing limit that prevents agents from engaging in Ponzi games. In the above expressions, C_t denotes consumption, p_t the relative price of trees in terms of consumption goods, e_t the number of trees owned by the representative household at the beginning of period t , and d_t the dividends per tree in period t . Dividends are

assumed to follow an exogenous stochastic process given by

$$d_{t+1} = e^{x_{t+1}} d_t,$$

where e^{x_t} denotes the gross growth rate of dividends. The natural logarithm of the gross growth rate of dividends, x_t , is assumed to follow an exogenous AR(1) process

$$x_{t+1} = (1 - \rho)\bar{x} + \rho x_t + \sigma \eta \varepsilon_{t+1},$$

and $\varepsilon_t \sim NIID(0, 1)$.¹⁴

The optimality conditions associated with the household's problem are the above budget constraint, the borrowing limit, and

$$p_t C_t^{\theta-1} = \beta E_t C_{t+1}^{\theta-1} (p_{t+1} + d_{t+1}).$$

Asset_model.m

In equilibrium we have that $C_t = d_t$ and $e_t = 1$.

$$p_t C_t^{\theta-1} = \beta E_t C_{t+1}^{\theta-1} (p_{t+1} + d_{t+1}).$$

$$x_{t+1} = (1 - \rho)\bar{x} + \rho x_t + \sigma \eta \varepsilon_{t+1},$$

```
13 %Define parameters
14 syms XBAR RHO THETA BETTA
15
16 %Define variables
17 syms p pp a ap
18
19 %Write equations (e1 and e2)
20 e1 = -p + BETTA * (1 + pp) * exp(THETA * ap);
21
22 e2 = -ap + (1-RHO) * XBAR + RHO * a;
23
24 %Create function f
25 f = [e1;e2];
26
```

2次近似と不動点による数値解の比較

Burnside (1998) shows that the non-explosive solution to this equation is of the form

$$y_t \equiv g(x_t, \sigma) = \sum_{i=1}^{\infty} \beta^i e^{a_i + b_i(x_t - \bar{x})}, \quad (9)$$

where

$$a_i = \theta \bar{x} i + \frac{\theta^2 \sigma^2 \eta^2}{2(1-\rho)^2} \left[i - \frac{2\rho(1-\rho^i)}{1-\rho} + \frac{\rho^2(1-\rho^{2i})}{1-\rho^2} \right]$$

and

$$b_i = \frac{\theta \rho(1-\rho^i)}{1-\rho}.$$

2次近似による数値解

A second-order approximation of (9) around $x_t = \bar{x}$ and $\sigma = 0$ yields

$$y_t \approx g(\bar{x}, 0) + g_x(x_t - \bar{x}) + \frac{1}{2}g_{xx}(x_t - \bar{x})^2 + \frac{1}{2}g_{\sigma\sigma}\sigma^2, \quad (10)$$

where

$$g_x = \frac{\theta\rho\beta e^{\theta\bar{x}}}{(1 - \beta e^{\theta\bar{x}})(1 - \beta e^{\theta\bar{x}}\rho)},$$
$$g_{xx} = \left(\frac{\rho\theta}{1 - \rho}\right)^2 \left[\frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}} - \frac{2\beta e^{\theta\bar{x}}\rho}{1 - \beta e^{\theta\bar{x}}\rho} + \frac{\beta e^{\theta\bar{x}}\rho^2}{1 - \beta e^{\theta\bar{x}}\rho^2} \right],$$

and

$$g_{\sigma\sigma} = \left(\frac{\theta\eta}{1 - \rho}\right)^2 \left[\frac{\beta e^{\theta\bar{x}}}{(1 - \beta e^{\theta\bar{x}})^2} + \left(\frac{\rho^2}{1 - \rho^2} - \frac{2\rho}{1 - \rho} \right) \right. \\ \left. \times \frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}} + \frac{2\rho^2}{1 - \rho} \frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}\rho} - \frac{\rho^4}{1 - \rho^2} \frac{\beta e^{\theta\bar{x}}}{1 - \beta e^{\theta\bar{x}}\rho^2} \right].$$

We follow Burnside (1998) and Collard and Juillard (2001b) and use the calibration $\beta = 0.95$, $\theta = -1.5$, $\rho = -0.139$, $\bar{x} = 0.0179$, and $\eta = 0.0348$. Then, evaluating the above expressions we obtain

$$y_t \approx 12.30 + 2.27(x_t - \bar{x}) + \frac{1}{2}0.42(x_t - \bar{x})^2 + \frac{1}{2}0.35\sigma^2.$$

, we have $f_0 = g(\bar{x}, 0) + \frac{1}{2} g_{\sigma\sigma} \sigma^2$, $f_1 = g_x$, and $f_2 = g_{xx}$.

Table 1

Second-order approximation to the policy function of the asset-pricing model

	f_0	f_1	f_2
<i>Benchmark calibration:</i>			
2nd-order approx.	12.48	2.27	0.42
Fixed-point algorithm	12.48	2.30	0.43
<i>High curvature, $\theta = -10$:</i>			
2nd-order approx.	4.79	4.83	6.07
Fixed-point algorithm	5.00	5.97	7.50
<i>High Persistence, $\rho = 0.9$:</i>			
2nd-order approx.	22.02	− 99.07	976.84
Fixed-point algorithm	14.50	−115.40	1137.81

Note: The benchmark calibration is $\beta = 0.95$, $\theta = -1.5$, $\rho = -0.139$, $\bar{x} = 0.0179$, $\sigma = 1$ and $\eta = 0.0348$. The rows labeled ‘2nd-order approx.’ are obtained either by evaluating Eq. (10) or by running the program `asset_run.m`. The rows labeled ‘Fixed-Point Algorithm’ are taken from Collard and Juillard (2001b), Table 2.

Table 1: 2nd Order approx.

THETA = -1.5

RHO = -0.139

f0 = 12.4788

f1 = 2.2731

f2 = 0.42053

Table 1: 2nd Order approx.

THETA = -10

RHO = -0.139

f0 = 4.7863

f1 = 4.8337

f2 = 6.0703

Table 1: 2nd Order approx.

THETA = -1.5

RHO = 0.9

f0 = 22.0183

f1 = -99.0732

f2 = 976.835

fx >>

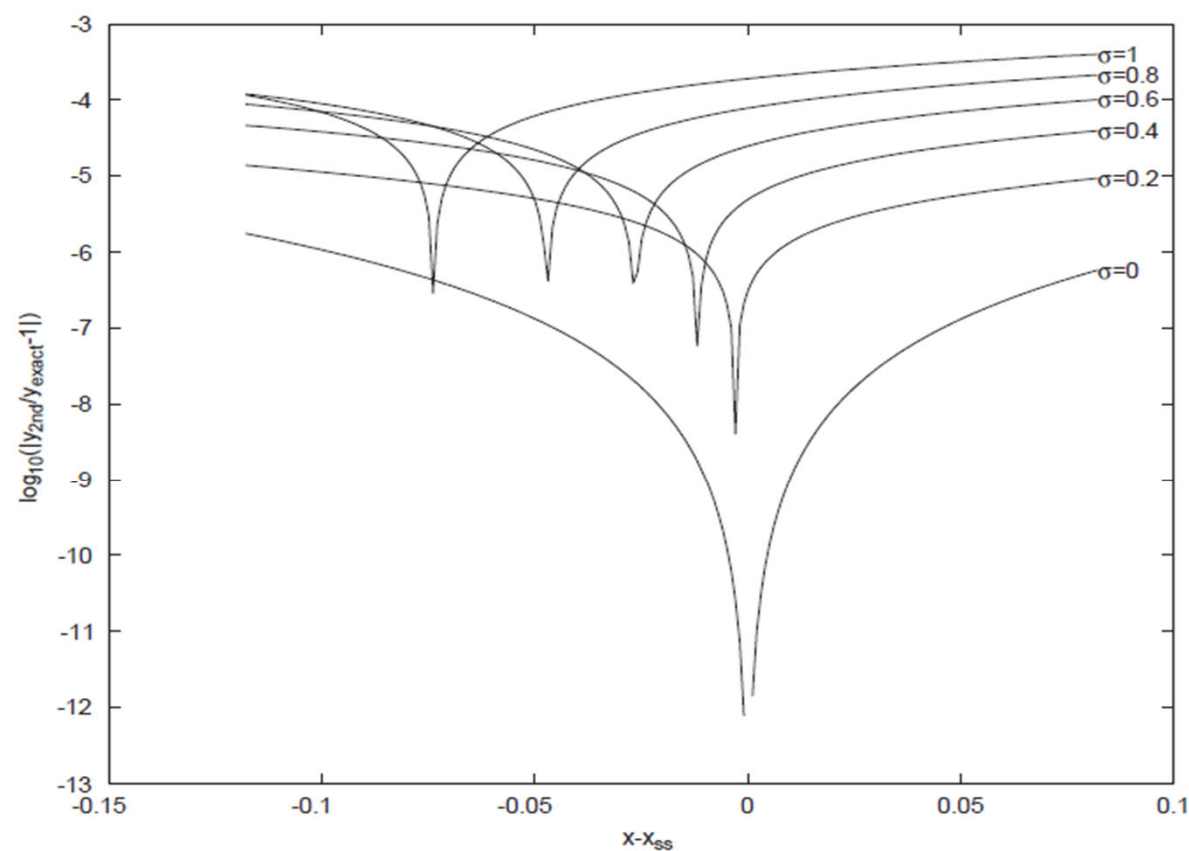


Fig. 1. Non-local accuracy test. Note: y denotes the price-dividend ratio, x (the state variable) denotes the log of the gross growth rate of dividends, σ denotes the parameter scaling the standard deviation of the exogenous shock, and x_{ss} denotes the steady-state value of x . All parameters take the values used in the benchmark calibration, as given in the note to Table 1. In computing the exact solution we truncated the infinite sum at $i = 1000$.

インパルス応答の1次近似と2次近似の比較

